

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

Fourth Semester of B. Tech. (CE) Examination

November 2012

CE 208 - Object Oriented Concepts & Visual C++ Programming

Date: 06.11.2012, Tuesday

Time: 01:30 p.m. to 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

Q - 1 Answer following questions:

1. How Class implement data abstraction? 01
2. What is main benefit of Operator Overloading? 01
3. How Inline Function is useful in programming? 01
4. Write syntax for constructor and destructor. 01
5. Explain *enum* keyword in C++. 01
6. State whether the following statement is True or False, give reason for same 01
"We can call friend function without creating instance of Class."
7. What is significance of Virtual function in C++? 01

Q - 2 Attempt following :

- (a) Why type conversion is necessary in C++? Write down difference between Implicit and Explicit type conversion in C++ compared with C. With example 06
- (b) What is polymorphism? Differentiate Compile time binding and run time binding. 04

OR

- (b) Is there any problem in hybrid Inheritance? If yes then Explain what the problem with its solution with example. If No then justify your answer. 04
- (c) Constructor will be overloaded or not? If yes then explain how many ways it's overloaded with example. If No then justify your answer. 04

OR

- (c) What is friend function? Write down its syntax in terms of declaration, definition and calling. 04

Q - 3 Attempt following :

- (a) Create two classes circle (radius, area) and rectangle (length, breadth, area). Write a C++ Program to create friend function "rest" to calculate area of dark portion. 06



- (b) Write a C++ Program to enter information of an employee and display it by using late binding technique. 04

OR

- (b) Differentiate Call by Value or Call by Reference by sorting example. 04
(c) What MFC stands? Explain WinMain function in detail. 04

OR

- (c) Write a program in C++ to overload arithmetic assignment operator (+=) as a member function. 04

SECTION – II

Q - 4 Attempt following :

1. What is the difference between Generalization and Aggregation? 02
2. The super-class has general characteristics that are _____ by all its sub-classes; the sub-classes have special characteristics that are _____ to each subclass. (Fill in the blanks) 01
3. Which OOAD model specifies the allowable scenarios that may occur for a system? 01
4. Differentiate Method and Message in terms of State Chart Diagram 02
5. Composition is a stronger form of which of the following? 01
 - a) Aggregation
 - b) Encapsulation
 - c) Inheritance
 - d) All of the above

Q - 5 Attempt following :

- (a) Differentiate Functional Modeling and Object Modeling in terms of Diagrams? 06
(b) When to use activity diagram? In which cases, activity diagram can't be used? 06
Draw an activity diagram of Online Exam System.

OR

- (b) Why DFD is not included in UML? What is SDLC? Explain it in terms of Object Oriented Designing. 06
(c) Explain the concept of Returning of Object with example. 02

OR

- (c) Explain the concept of Copy Constructor with example. 02

Q - 6 Attempt following :

- (a) Create a class hierarchy to organize the following drink classes: 03
Mineral water, wine, alcoholic, nonalcoholic, grape juice, soda, bears.
(b) Draw use case diagram for Library Management System (LMS). 05

OR

- (b) Draw class diagram for Hospital Management System. 05
(c) Write a program that illustrates the hierarchical inheritance in creating two new class called savings account and current account from the base class account. 06

OR

(c) Answer the Following with reason:

- 1) Which of the following statement is true about preprocessor directives?
 - a) These are lines read and processed by the preprocessor
 - b) They do not produce any code by themselves
 - c) These must be written on their own line
 - d) They end with a semicolon
- 2) An Inline function involves some additional overhead in running time.
 - a) True
 - b) False
- 3) Observe following function declaration and choose the best answer:
int divide (int a, int b = 2)
 - a) Variable b is of integer type and will always have value 2
 - b) Variable a and b are of int type and the initial value of both variables is 2
 - c) Variable b will have value 2 if not specified when calling function
 - d) Variable b is international scope and will have value 2
